



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

SEMESTER: VII

**DEPARTMENT
OF
COMPUTER SCIENCE AND ENGINEERING**

PRACTICE WORK BOOK

SUBJECT CODE: 3170724

SUBJECT NAME: MACHINE LEARNING

Subject Code:	3170724	Subject Name	Credit:	4
Semester:	VII	Machine Learning	Hours:	5
Department:	Computer Science and Engineering			

CHAPTER: 1**INTRODUCTION TO MACHINE LEARNING**

Overview of Human Learning and Machine Learning, Types of Machine Learning, Applications of Machine Learning, Tools and Technology for Machine Learning.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Give the difference between supervised learning and unsupervised learning. (Dec-2021) [NLJIET]	3	CO1	L1	RM
2	Define Machine learning? Briefly explain the types of learning. (Jun-2022) [NLJIET]	3	CO1	L2	UN
3	Define issues in machine Learning. (Jun-2022) [NLJIET]	3	CO1	L1	RM
4	Define: a. Supervised Learning b. Classification c. Regression (Jun-2022) [NLJIET]	3	CO1	L1	RM
5	Define Machine learning and list out few applications in Engineering? (Jan-2023) [NLJIET]	3	CO1	L1	RM
6	Compare Different types of Machine Learning. (Jun-2023) [NLJIET]	3	CO1	L4	AN
7	Explain the concept of penalty and reward in reinforcement learning. (Jun-2023) [NLJIET]	3	CO1	L2	UN
8	Differentiate Human Learning & Machine Learning. (Dec-2023) [NLJIET]	3	CO1	L1	RM
9	List out different issues in machine Learning. (Dec-2023) [NLJIET]	3	CO1	L1	RM
10	What is reinforcement learning? State key features of reinforcement learning. (May-2024) (Nov-2025) [NLJIET]	3	CO1	L2	UN
11	What is learning? What are the types of learning? (Nov-2024) [NLJIET]	3	CO1	L2	UN
12	What are the applications of unsupervised learning? (Nov-	3	CO1	L2	UN

	2024) [NLJIET]				
13	What is machine learning? What are key tasks of machine learning? (May-2025) [NLJIET]	3	CO1	L2	UN
14	Define Machine Learning & Human Learning. Give at least two differences between machine learning and human learning. (Nov-2025) [NLJIET]	3	CO1	L2	UN
15	Can all problems be solved using machine learning? Explain your response in detail. (Nov-2025) [NLJIET]	3	CO1	L2	UN
16	What is well-posed learning problem? (May-2026) [NLJIET]	3	CO1	L2	UN
17	How do machine learn? (Nov-2024) [NLJIET]	4	CO1	L2	UN
18	Explain the concept of penalty and reward in reinforcement learning. (Dec-2021) (Jun-2022) [NLJIET]	4	CO1	L2	UN
19	Distinguish lazy vs eager learner with an example. (Jan-2023) [NLJIET]	4	CO1	L1	RM
20	What is Machine Learning? List out its applications and possible ethical issues of machine learning applications? (Jun-2023) [NLJIET]	4	CO1	L1	RM
21	Compare the different types of machine learning. (May-2024) [NLJIET]	4	CO1	L4	AN
22	What do you mean by a well-posed learning problem? Explain important features that are required to well-define a learning problem. (May-2025) [NLJIET]	4	CO1	L2	UN
23	How does reinforcement learning differ from supervised learning, and in what types of applications is reinforcement 04 learning particularly useful? (May-2026) [NLJIET]	4	CO1	L2	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	What do you mean by a well-posed learning problem? Explain important features that are required to well-define a learning problem. (Dec-2021) (Jun-2022) (Jun-2023) [NLJIET]	7	CO1	L2	UN
2	Draw and explain the flow diagram of machine learning	7	CO1	L2	UN

	procedure. (Jun-2022) [NLJIET]				
3	List and explain the types of machine learning in brief. (Jun-2022) [NLJIET]	7	CO1	L2	AN
4	Explain Reinforcement Learning in detail. (Dec-2023) [NLJIET]	7	CO1	L2	UN
5	Organize the different types of unsupervised learning and show their differences with a sample application in each area. [NLJIET]	7	CO1	L3	AP
6	Explain Supervised Learning in detail. (Dec-2023) [NLJIET]	7	CO1	L2	UN
7	Write a short note on: lazy vs. eager learner. (May-2025) [NLJIET]	7	CO1	L2	UN
8	Distinguish between supervised learning, semi-supervised learning, and unsupervised learning. (May-2025) [NLJIET]	7	CO1	L2	UN

CHAPTER: 2

PREPARING TO MODEL

Machine Learning activities, Types of data in Machine Learning, Structures of data, Data quality and remediation, Data Pre-Processing: Dimensionality reduction, Feature subset selection.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	How can we take care of outliers in data? (Dec-2021) (Jun-2022) [NLJIET]	3	CO1	L3	AP
2	What are the basic data types in machine learning? Give an example of each one of them. (Jun-2023) [NLJIET]	3	CO1	L1	RM
3	What are the different techniques for data pre-processing? Explain in brief (Jun-2023) [NLJIET]	3	CO1	L2	UN
4	What is the difference between Qualitative Data and Quantitative Data? (Dec-2023) [NLJIET]	3	CO1	L2	UN
5	How does discretization help us to deal with outliers? (May-2024) [NLJIET]	3	CO1	L3	AP
6	What is IQR? How is it measured? (Nov-2024) [NLJIET]	3	CO1	L2	UN

7	What are different shapes of histogram? What are 'bins'? (May-2025) [NLJIET]	3	CO1	L2	UN
8	Define outliers in data. How to handle them? (May-2026) [NLJIET]	3	CO1	L3	AP
9	What is categorical data? Explain its types with examples. (Jan-2023) [NLJIET]	4	CO1	L2	UN
10	What is outlier? How can we take care of outlier? (Jun-2023) [NLJIET]	4	CO1	L2	UN
11	Explain Standard Deviation and Variance with an example in machine learning. (Dec-2023) [NLJIET]	4	CO1	L2	UN
12	Explain qualitative and quantitative data. Differentiate between the two. (May-2024) [NLJIET]	4	CO1	L2	UN
13	Explain, in details, the different strategies of addressing missing data values. (May-2025) [NLJIET]	4	CO1	L2	UN
14	Differentiate central tendency and data dispersion with simple examples. (May-2026) [NLJIET]	4	CO1	L4	AN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	What are the Techniques Provided in Data Preprocessing? Explain in brief. (Jun-2022) (Dec-2023) [NLJIET]	7	CO1	L2	UN
2	Explain dimensionality reduction with its advantages and disadvantages. (Dec-2023) [NLJIET]	7	CO1	L2	UN
3	What is IQR? How is it measured? Explain with proper example. (May-2024) [NLJIET]	7	CO1	L3	AP
4	Design a simple data set with sample records, including at least one attribute of each different data type commonly used in machine learning.[NLJIET]	7	CO1	L6	CR
5	Give the attribute of Descriptive statistics in ML Explain with proper example. [NLJIET]	7	CO1	L2	UN
6	Explain different ways of exploring categorical data that	7	CO1	L2	UN

	integrate with proper examples. [NLJIET]				
7	Explain the different strategies for addressing missing data values in detail. [NLJIET]	7	CO1	L2	UN
8	Compare Scatter plot, Histogram and Box plot. [NLJIET]	7	CO1	L4	AN
9	Explain in details, the different strategies of addressing missing data values? (Nov-2024) [NLJIET]	7	CO1	L2	UN
10	Explain qualitative and quantitative data in details. Differentiate between the two. (May-2025) [NLJIET]	7	CO1	L2	UN
11	State the different techniques for data pre-processing? Explain, in brief, dimensionality reduction and feature selection. (Nov-2025) [NLJIET]	7	CO1	L2	UN
12	Explain steps and activities to use machine learning for any real word application. (May-2026) [NLJIET]	7	CO1	L2	UN
13	Discuss different types of data in machine learning. (May-2026) [NLJIET]	7	CO1	L2	UN

CHAPTER: 3**MODELLING AND EVALUATION**

Selecting a Model: Predictive/Descriptive, Training a Model for supervised learning, model representation and interpretability, Evaluating performance of a model, Improving performance of a model.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Explain the training of Predictive Model. (Jun-2022) [NLJIET]	3	CO1	L2	UN
2	What is K-fold cross validation? (Dec-2023) [NLJIET]	3	CO1	L2	UN
3	What are predictive models? Explain with the help of an example. (Nov-2025) [NLJIET]	3	CO1	L2	UN
4	What are descriptive models? Explain with the help of an example. (Nov-2025) [NLJIET]	3	CO1	L2	UN
5	Consider the following confusion matrix of the win/loss prediction of cricket match. Calculate model accuracy and	4	CO1	L3	AP

	error rate for the same. (Dec-2021) [NLJIET]													
	<table><tr><td></td><td>Actual Win</td><td>Actual Loss</td></tr><tr><td>Predicted Win</td><td>82</td><td>7</td></tr><tr><td>Predicted Loss</td><td>3</td><td>8</td></tr></table>		Actual Win	Actual Loss	Predicted Win	82	7	Predicted Loss	3	8				
	Actual Win	Actual Loss												
Predicted Win	82	7												
Predicted Loss	3	8												
6	Explain the process of K-fold-cross-validation method. (Jun-2023) [NLJIET]	4	CO1	L2	UN									
7	What is sampling? Explain Bootstrap sampling. (Jun-2023) [NLJIET]	4	CO1	L2	UN									
8	Explain Under-fitting and Over-fitting with an example. (Dec-2023) [NLJIET]	4	CO1	L2	UN									
9	Define sampling with replacement. Give proper example. (May-2024) [NLJIET]	4	CO1	L1	RM									
10	What is sampling without replacement? Give proper example. (May-2024) [NLJIET]	4	CO1	L3	AP									
11	Explain Holdout method. (Nov-2024) [NLJIET]	4	CO1	L2	UN									

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Explain K-fold cross validation method with suitable example. (Dec-2021) [NLJIET]	7	CO1	L2	UN
2	List the methods for Model evaluation. Explain each. How we can improve the performance of model. (Jun-2022) [NLJIET]	7	CO1	L1	RM
3	Elaborate the cross validation in training a model. (Jan-2023) [NLJIET]	7	CO1	L2	UN
4	What is model accuracy in reference to classification? Also Explain the performance parameters Precision, Recall and F-measure with its formula and example. (Jan-2023) [NLJIET]	7	CO1	L2	UN
5	Consider the following confusion matrix of the win/loss prediction of cricket match. Calculate model accuracy and error rate, sensitivity, precision, F-measure and kappa value for the same. (Jun-2023) [NLJIET]	7	CO1	L3	AP

		Actual Win	Actual Loss				
	Predicted Win	85	4				
	Predicted Loss	2	9				
6	While predicting malignancy of tumor of a set of patients using a classification model, following are the data recorded: (a) Correct predictions – 15 malignant, 75 benign (b) Incorrect predictions – 3 malignant, 7 benign Calculate the model accuracy, error rate, Kappa value, sensitivity, precision, and F measure of the model. (Jun-2023) [NLJIET]	7	CO1	L3	AP		
7	What are overfitting and underfitting in context of machine learning models? What is the major cause of overfitting and underfitting? (May-2024) [NLJIET]	7	CO1	L2	UN		
8	How to improve the performance of a learning model? Explain in details. (May-2024) [NLJIET]	7	CO1	L2	UN		
9	What are predictive models? What are descriptive models? Give examples of both types of models. Explain the difference between these types of models. (Nov-2024) [NLJIET]	7	CO1	L2	UN		
10	How would you evaluate the success of an unsupervised learning model? What are the most popular measures of performance for an unsupervised learning model? (Nov-2024) [NLJIET]	7	CO1	L5	EL		
11	Write a short note on: holdout method. (May-2025) [NLJIET]	7	CO1	L2	UN		
12	How would you evaluate the success of an unsupervised learning model? What are the most popular measures of performance for an unsupervised learning model? Explain any two with example. (Nov-2025) [NLJIET]	7	CO1	L2	UN		
13	What is feature selection in context of machine learning? Why is it needed? List the different approaches of feature selection? Explain any one with example. (Nov-2025) [NLJIET]	7	CO1	L2	UN		
14	Discuss holdout and k-fold cross validation for supervised learning. (May-2026) [NLJIET]	7	CO1	L2	UN		

CHAPTER: 4**BASICS OF FEATURE ENGINEERING**

Feature and Feature Engineering, Feature transformation: Construction and extraction, Feature subset selection : Issues in high-dimensional data, key drivers, measure and overall process

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Explain the need of feature engineering in ML. (Jun-2022) [NLJIET]	3	CO2	L2	UN
2	What is feature engineering? (May-2024) (Nov-2024) [NLJIET]	3	CO2	L2	UN
3	When can a feature be termed as irrelevant? How can it be measured? (May-2024) (Nov-2024) [NLJIET]	3	CO2	L3	AP
4	What is principal component analysis? (Nov-2024) [NLJIET]	3	CO2	L2	UN
5	What is a feature? Explain with an example. (May-2025) [NLJIET]	3	CO2	L2	UN
6	Differentiate feature transformation with feature selection. (May-2025) [NLJIET]	3	CO2	L4	AN
7	Differentiate PCA and LDA. (Jun-2022) [NLJIET]	4	CO2	L4	AN
8	What is the purpose of Singular value decomposition? How does it achieve? (Jan-2023) [NLJIET]	4	CO2	L3	AP
9	Show various distance-based similarity measure with its example. (Jan-2023) [NLJIET]	4	CO2	L2	UN
10	What is Feature Subset Selection? Explain it in detail. (Dec-2023) [NLJIET]	4	CO2	L2	UN
11	Explain the different situations that necessitate feature construction. [NLJIET]	4	CO2	L2	UN
12	When can a feature be termed as redundant? Give the measures to determine the potentially redundant features. [NLJIET]	4	CO2	L4	AN
13	Compare Euclidean distance with Manhattan distance? [NLJIET]	4	CO2	L4	AN
14	Explain the feature selection approaches. (Nov-2024)	4	CO2	L2	UN

	[NLJIET]				
15	Differentiate feature construction and feature extraction. (May-2026) [NLJIET]	4	CO2	L4	AN
16	Differentiate feature relevance and feature redundancy. (May-2026) [NLJIET]	4	CO2	L4	AN
Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Explain SVD as a feature extraction technique with suitable example. (Dec-2021) [NLJIET]	7	CO2	L2	UN
2	What is principal component analysis? How does it work? Explain. (Jan-2023) [NLJIET]	7	CO2	L2	UN
3	Define feature and explain the process of transforming numeric features to categorical features with suitable example. (Jan-2023) [NLJIET]	7	CO2	L2	UN
4	Explain with an example, main underlying concept of feature extraction. What are the most popular algorithms of feature extraction, briefly explain any one. (Jun-2023) [NLJIET]	7	CO2	L2	UN
5	What is feature selection? Why it is needed? What are the different approaches of feature selection, briefly explain any one. (Jun-2023) [NLJIET]	7	CO2	L2	UN
6	Write a Short note of PCA Algorithm with examples.[NLJIET]	7	CO2	L2	UN
7	Explain the wrapper approach of feature selection. What are the merits and demerits of this approach? [NLJIET]	7	CO2	L2	UN
8	What is feature selection? Why is it needed? Explain the different approaches of feature selection? [NLJIET]	7	CO2	L2	UN

Exploring Emerging Technologies

CHAPTER: 5 OVERVIEW OF PROBABILITY

Statistical tools in Machine Learning, Concepts of probability, Random variables, Discrete distributions, Continuous distributions, Multiple random variables, Central limit theorem, Sampling distributions, Hypothesis testing, Monte Carlo Approximation.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective. (Dec-2021) [NLJIET]	3	CO2	L3	AP
2	In a communication system each data packet consists of 1000 bits. Due to the noise, each bit may be received in error with probability 0.1. It is assumed bit errors occur independently. Find the probability that there are more than 120 errors in a certain data packet. (Dec-2021) [NLJIET]	3	CO2	L3	AP
3	What is conditional probability? Define its importance. (Jan-2023) [NLJIET]	3	CO2	L1	RM
4	Define the following terms. (i) Variance (ii) Covariance (iii) Joint Probability (Jan-2023) [NLJIET]	3	CO2	L2	UN
5	What is Bernoulli distribution? Explain briefly with its formula. (Jan-2023) [NLJIET]	3	CO2	L2	UN
6	What is joint probability? What is its formula? (Jun-2023) [NLJIET]	3	CO2	L1	RM
7	Define probability of union of two events with equation. (Jun-2023) (Nov-2025) [NLJIET]	3	CO2	L1	RM
8	What is Monte Carlo Approximation? (Dec-2023) [NLJIET]	3	CO2	L2	UN
9	Define central limit theorem. (May-2025) (May-2026) [NLJIET]	3	CO2	L1	RM
10	Explain Key elements of Machine Learning. Explain various function approximation methods. (Jun-2022) [NLJIET]	4	CO2	L2	UN
11	What is data sampling? Explain data sampling methods? (Jun-2022) [NLJIET]	4	CO2	L2	UN
12	Explain Binomial Distribution with an example. (Jun-2022) [NLJIET]	4	CO2	L2	UN

13	Explain Hypothesis testing with basic assumptions. (Dec-2023) [NLJIET]	4	CO2	L2	UN
14	What is conditional probability means? What is the formula of it? (May-2025) [NLJIET]	4	CO2	L1	RM
15	What is Poisson distribution? What is the formula? (May-2025) [NLJIET]	4	CO2	L1	RM

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Explain Monte Carlo Approximation. (Jun-2022) [NLJIET]	7	CO2	L2	UN
2	Explain Probability Distribution Function and Cumulative Distribution Function with an Example. (Dec-2023) [NLJIET]	7	CO2	L2	UN
3	Explain Sampling Distribution with different types of Sampling. (Dec-2023) [NLJIET]	7	CO2	L2	UN
4	Explain common discrete distributions. (Nov-2024) [NLJIET]	7	CO2	L2	UN
5	Discuss parametric distributions defined on discrete sample space. (May-2026) [NLJIET]	7	CO2	L2	UN
6	Discuss hypothesis testing in detail. (May-2026) [NLJIET]	7	CO2	L2	UN

CHAPTER: 6**BAYESIAN CONCEPT LEARNING**

Importance of Bayesian methods, Bayesian theorem, Bayes' theorem and concept learning, Bayesian Belief Network

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	What is likelihood probability? Give an example. (Jun-2022) [NLJIET]	3	CO5	L1	RM
2	Explain posterior probability with its formula. (Jan-2023) [NLJIET]	3	CO5	L2	UN
3	Define Bayes theorem. (May-2026) [NLJIET]	3	CO5	L1	RM

4	Explain how Naïve Bayes classifier is used for Spam Filtering. (Dec-2021) [NLJIET]	4	CO5	L2	UN
5	Explain the concept of Bayesian belief network. (Jan-2023) [NLJIET]	4	CO5	L2	UN
6	What is likelihood probability? Give an example. (Dec-2021) (Jun-2023) [NLJIET]	4	CO5	L1	RM
7	Explain likelihood probability with an example	4	CO5	L2	AN
8	What is concept learning? Explain with example (Jun-2023) [NLJIET]	4	CO5	L2	UN
9	What are Bayesian Belief networks? Where are they used? Can they solve all types of problems? (Nov-2025) [NLJIET]	4	CO5	L2	UN
10	What is likelihood probability? Explain with an example. (Nov-2025) [NLJIET]	4	CO5	L2	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Explain Bayes' theorem in details. (Jun-2022) [NLJIET]	7	CO5	L2	UN
2	Explain Bayes' theorem in details with the use of Bayesian classifiers. (Dec-2023) [NLJIET]	7	CO5	L2	UN
3	Explain naïve bayes classifier with an example. (May-2024) [NLJIET]	7	CO5	L2	UN
4	Explain Bayesian belief network. (May-2024) [NLJIET]	7	CO5	L2	UN
5	What are Bayesian belief networks? Where are they used? Can they solve all types of problems? (Nov-2024) (May-2025) [NLJIET]	7	CO5	L2	UN
6	Explain Naïve Bayes classifier with an example of its use in practical life. (Nov-2025) [NLJIET]	7	CO5	L2	UN
7	Explain Naïve Bayes classifier with a simple example. (May-2026) [NLJIET]	7	CO5	L2	UN

CHAPTER: 7
SUPERVISED LEARNING: CLASSIFICATION AND REGRESSION

Supervised Learning, Classification Model, Learning steps, Classification algorithms, Regression, Regression algorithms,

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Explain sum of squares due to error in multiple linear regression with example. (Dec-2021) [NLJIET]	3	CO3	L2	UN
2	Explain dependent variable and an independent variable in a linear equation with example. (Dec-2021) (Jun-2023) [NLJIET]	3	CO3	L2	UN
3	What are the conditions of a negative slope in linear regression? (Dec-2021) [NLJIET]	3	CO3	L2	UN
4	Write the strength and weakness of decision tree method. (Jun-2023) [NLJIET]	3	CO3	L1	RM
5	What is Multiple Linear regression? (Dec-2023) [NLJIET]	3	CO3	L2	UN
6	Explain the classification learning steps. (May-2024) [NLJIET]	3	CO3	L2	UN
7	Why the kNN algorithm is called a lazy learner? (May-2024) [NLJIET]	3	CO3	L2	UN
8	Differentiate between supervised learning versus unsupervised learning. (May-2024) [NLJIET]	3	CO3	L4	AN
9	What are the strengths and weaknesses of SVM? (Jan-2023) [NLJIET]	3	CO3	L2	UN
10	Explain the classification learning steps. (Nov-2024) [NLJIET]	3	CO3	L2	UN
11	What is the cost of misclassification in SVM? (Nov-2024) (May-2025) [NLJIET]	3	CO3	L2	UN
12	What is under-fitting in context of machine learning models? State its major causes. (Nov-2025) [NLJIET]	3	CO3	L2	UN
13	Explain curve linear negative slope and curve linear positive slope in a graph. (Nov-2025) [NLJIET]	3	CO3	L2	UN
14	Define simple linear regression using a graph. Explain slope and intercept. (Nov-2025) [NLJIET]	3	CO3	L2	UN

15	How to avoid overfitting in decision tree based classifier? (May-2026) [NLJIET]	3	CO3	L2	UN
16	How to identify correct hyperplane in SVM? (May-2026) [NLJIET]	3	CO3	L2	UN
17	Write a note on KNN. (Jun-2022) [NLJIET]	4	CO3	L2	UN
18	List Classification algorithms. Explain Decision Tree as classification method. (Jun-2022) [NLJIET]	4	CO3	L2	UN
19	Define Entropy. Show its importance with suitable example. (Jan-2023) [NLJIET]	4	CO3	L2	UN
20	Define linear regression. Also explain Sum of squares with its formula. (Jan-2023) [NLJIET]	4	CO3	L2	UN
21	Explain Decision tree algorithm. (Jun-2023) [NLJIET]	4	CO3	L2	UN
22	What is a bad decision boundary in SVM? Explain it in brief. (Dec-2023) [NLJIET]	4	CO3	L2	UN
23	What are the factors determining the effectiveness of SVM? (May-2024) (Dec-2021) (May-2025) [NLJIET]	4	CO3	L2	UN
24	Describe the strengths and weaknesses of decision tree. (May-2024) [NLJIET]	4	CO3	L2	UN
25	Explain logistic regression. (Nov-2024) [NLJIET]	4	CO3	L2	UN
26	Explain ridge regression. (Nov-2024) [NLJIET]	4	CO3	L2	UN
27	What are the advantages of the kNN algorithm? (May-2025) [NLJIET]	4	CO3	L2	UN
28	What are the disadvantages of the kNN algorithm? (May-2025) [NLJIET]	4	CO3	L2	UN
29	What are the different types of supervised learning? Explain them with a sample application in each area. (Nov-2025) [NLJIET]	4	CO3	L2	UN
30	What is decision tree? What are the different types of nodes? Explain with example. (Nov-2025) [NLJIET]	4	CO3	L2	UN
31	Differentiate classification and regression. (May-2026) [NLJIET]	4	CO3	L4	AN
32	How to improve accuracy of a linear regression model? (May-	4	CO3	L4	AN

	2026) [NLJIET]				
33	How to measure accuracy of a classification model? (May-2026) [NLJIET]	4	CO3	L4	AN
DESCRIPTIVE QUESTIONS					
Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Discuss appropriate problems for decision tree learning in detail. (Dec-2021) [NLJIET]	7	CO3	L2	UN
2	Discuss the error rate and validation error in the kNN algorithm. (Dec-2021) [NLJIET]	7	CO3	L2	UN
3	Explain the process of Supervised Learning Model. (Jun-2022) [NLJIET]	7	CO3	L2	UN
4	Explain KNN algorithm with suitable example. (Jan-2023) [NLJIET]	7	CO3	L2	UN
5	Explain decision tree approach with suitable example. (Jan-2023) [NLJIET]	7	CO3	L2	UN
6	What is supervised learning? Draw and explain classification steps in detail. (Jun-2023) [NLJIET]	7	CO3	L2	UN
7	Explain Decision Tree in details. (Dec-2023) [NLJIET]	7	CO3	L2	UN
8	Explain k-Nearest Neighbor classification method in detail with its advantages and disadvantages. (Dec-2023) [NLJIET]	7	CO3	L2	UN
9	Discuss the decision tree algorithm in detail. (May-2024) [NLJIET]	7	CO3	L2	UN
10	Explain the difference between linear and logistics regression with example. (May-2024) [NLJIET]	7	CO3	L2	UN
11	Discuss KNN model in detail. (Nov-2024) [NLJIET]	7	CO3	L2	UN
12	Discuss decision tree algorithm in detail. (Nov-2024) [NLJIET]	7	CO3	L2	UN
13	What is regression? Differentiate between logistic and linear regression with suitable example. (May-2025) [NLJIET]	7	CO3	L2	UN
14	What is overfitting in decision tree? Explain the technique to avoid the overfitting. (May-2025) [NLJIET]	7	CO3	L2	UN

15	Discuss the kNN model in detail. (Nov-2025) [NLJIET]	7	CO3	L2	UN
16	Write a short note on: SVM. (Nov-2025) [NLJIET]	7	CO3	L2	UN
17	Explain decision tree based classifier with a simple example. (May-2026) [NLJIET]	7	CO3	L2	UN

CHAPTER: 8 UNSUPERVISED LEARNING

Supervised vs. Unsupervised Learning, Applications, Clustering, Association rules

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Mention few applications areas of unsupervised learning in Engineering. (Jan-2023) [NLJIET]	3	CO4	L1	RM
2	Discuss the strengths and weaknesses of the k-means algorithm. (May-2024) [NLJIET]	3	CO4	L2	UN
3	What are different clustering techniques? (May-2025) [NLJIET]	3	CO4	L1	RM
4	Association rule mining is probabilistic method of unsupervised learning. True or False? Justify your answer. (May-2026) [NLJIET]	3	CO4	L2	UN
5	Describe the concept of single link and complete link in the context of hierarchical clustering. (Dec-2021) [NLJIET]	4	CO4	L2	UN
6	Describe the main difference in the approach of k-means and k-medoids algorithms with a neat diagram. (Dec-2021) [NLJIET]	4	CO4	L2	UN
7	Briefly explain K-Medoids. (Jan-2023) [NLJIET]	4	CO4	L2	UN
8	Explain K-mean clustering algorithm. (Jun-2023) [NLJIET]	4	CO4	L2	UN
9	What is Clustering in Unsupervised Learning? Explain it in detail. (Dec-2023) [NLJIET]	4	CO4	L2	UN
10	Explain Apriori algorithm with its Limitation. (Dec-2023) [NLJIET]	4	CO4	L2	UN
11	Explain hierarchical clustering. (May-2024) (Nov-2024) [NLJIET]	4	CO4	L2	UN

12	What is a dendrogram? Explain its use. (Nov-2024) (May-2025) [NLJIET]	4	CO4	L2	UN
13	Explain density-based methods. (May-2024) (Nov-2025) [NLJIET]	4	CO4	L2	UN
14	Explain about Hierarchical clustering algorithm. (May-2025) [NLJIET]	4	CO4	L2	UN
15	Explain about EM algorithm. (May-2025) [NLJIET]	4	CO4	L2	UN
16	What are the different types of unsupervised learning? Explain them with a sample application in each area. (Nov-2025) [NLJIET]	4	CO4	L2	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Explain how the Market Basket Analysis uses the concepts of association analysis. (Dec-2021) [NLJIET]	7	CO4	L2	UN
2	Explain the Apriori algorithm for association rule learning with an example. (Dec-2021) (Jun-2023) [NLJIET]	7	CO4	L2	UN
3	What is Clustering? Explain K-mean clustering algorithm. (Jun-2022) [NLJIET]	7	CO4	L2	UN
4	How does the apriori principle help in reducing the calculation overhead for a market basket analysis? Explain with an example. (Jan-2023) (Nov-2025) [NLJIET]	7	CO4	L2	UN
5	Explain k-means clustering technique. (Jan-2023) [NLJIET]	7	CO4	L2	UN
6	Explain density-based methods – DBSCAN. (Nov-2024) [NLJIET]	7	CO4	L2	UN
7	Compare various clustering algorithms along with their advantages and limitations. (May-2026) [NLJIET]	7	CO4	L4	AN

CHAPTER: 9**NEURAL NETWORK**

Introduction to neural network, Biological and Artificial Neurons, Types of Activation functions, Implementation of ANN, Architecture, Learning process, Backpropagation, Deep Learning

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	What is difference between Machine Learning and Deep Learning (Jun-2022) [NLJIET]	3	CO3	L2	UN
2	Briefly explain Perceptron and Mention its limitation. (Jan-2023) [NLJIET]	3	CO3	L2	UN
3	What is Deep Learning? (Dec-2023) [NLJIET]	3	CO3	L2	UN
4	Explain Forward pass and Backward pass in neural network. (Dec-2023) [NLJIET]	3	CO3	L2	UN
5	What is deep learning? Explain in brief with an example. (Nov-2025) [NLJIET]	3	CO3	L2	UN
6	A significant increase in the number of hidden layers in a neural network is more likely to cause underfitting. True or False? Justify your answer. (May-2026) [NLJIET]	3	CO3	L2	UN
7	Explain Rosenblatt's perceptron model. (Dec-2021) [NLJIET]	4	CO3	L2	UN
8	Draw a flow chart which represents backpropagation algorithm. (Dec-2021) [NLJIET]	4	CO3	L2	UN
9	Write a short note on feed forward neural network. (Jun-2022) [NLJIET]	4	CO3	L2	UN
10	List out the characteristics of Artificial Neural Network and explain it in brief. (Dec-2023) [NLJIET]	4	CO3	L2	UN
11	Explain the different activation function used in neural network? (Nov-2024) [NLJIET]	4	CO3	L2	UN
12	Explain the ADALINE network model. (Nov-2024) [NLJIET]	4	CO3	L2	UN
13	In which applications does using deep learning provide significant performance improvements compared to traditional machine learning methods? (May-2026) [NLJIET]	4	CO3	L4	AN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	*L1-L6	RBT Level
1	Describe, in details, the process of adjusting the	7	CO3	L4	AN

	interconnection weights in a multi-layer neural network. (Dec-2021) [NLJIET]				
2	Explain, with example, the challenge in assigning synaptic weights for the interconnection between neurons? How can this challenge be addressed? (Dec-2021) [NLJIET]	7	CO3	L2	UN
3	Show the Step, ReLU and sigmoid activation functions with its equations and sketch. (Jan-2023) [NLJIET]	7	CO3	L2	UN
4	Explain in detail, the backpropagation algorithm. What are the limitations of this algorithm? (Jun-2023) [NLJIET]	7	CO3	L2	UN
5	Explain the backpropagation algorithm. What are the limitations of this algorithm? (May-2024) [NLJIET]	7	CO3	L2	UN
6	What are the different types of activation functions popularly used? Explain each of them. (Jun-2023) [NLJIET]	7	CO3	L2	UN
7	Explain the different types of activation functions used in neural network. (May-2024) [NLJIET]	7	CO3	L2	UN
8	Explain the basic structure of a multi-layer perceptron. Explain how it can solve the XOR problem. (May-2025) [NLJIET]	7	CO3	L2	UN
9	Explain, in details, the backpropagation algorithm. What are the limitations of this algorithm? (May-2025) [NLJIET]	7	CO3	L2	UN
10	Explain the learning process of an ANN. Explain, with example, the challenge in assigning synaptic weights for the interconnection between neurons? How can this challenge be addressed? (Nov-2025) [NLJIET]	7	CO3	L2	UN
11	Describe the structure of an artificial neuron. How is it similar to a biological neuron? Explain its main components? (Nov-2025) [NLJIET]	7	CO3	L2	UN
12	Compare various activation functions for neural networks along with their advantages and limitations. (May-2026) [NLJIET]	7	CO3	L4	AN

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